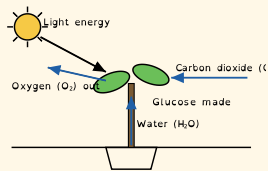


What is photosynthesis?

Photosynthesis is how green plants make their own food. They use light energy to turn carbon dioxide and water into glucose and oxygen, inside the chloroplasts. It is endothermic (it takes in energy).



Inputs and outputs of photosynthesis

Word & symbol equation

carbon dioxide + water $\rightarrow$ (light + chlorophyll) $\rightarrow$ glucose + oxygen

**Key vocabulary**

- **Reactants** — what you start with: carbon dioxide + water
- **Products** — what is made: glucose + oxygen
- **Chlorophyll** — green pigment that traps light energy
- **Endothermic** — a reaction that takes IN energy
- **Limiting factor** — the thing in short supply that slows the rate

Common mistakes

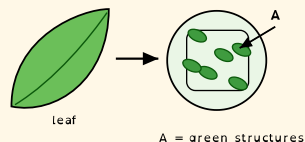
- Swapping the gases. The plant takes IN carbon dioxide and gives OUT oxygen — for example, the bubbles from pondweed are oxygen.
- Calling light a raw material. The raw materials are water and carbon dioxide; light is the energy.
- Saying 'in the leaf'. Be exact — it happens in the chloroplast inside the leaf cell.

Exam tip

Remember: light is the ENERGY source, not a raw material. The two raw materials (reactants) are carbon dioxide and water.

Diagram question — Q5

Look at the diagram. (a) Name structure A. (b) Name the green pigment it holds.

**Quick questions**

1. State the two reactants (raw materials) for photosynthesis.

2. Name the two products of photosynthesis.

3. Photosynthesis is endothermic. What does 'endothermic' mean?

4. Give two ways a plant uses the glucose it makes.

Apply it — Q6

A grower gives a plant more light, but the rate of photosynthesis does not increase. Suggest why.

Sentence starter: The rate does not increase because another factor (____ or ____) is now the limiting factor.
